HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · REAL ESTATE · PROJECTS & DESIGN

STEALTH SWING SHOWER ENCLOSURE · 16 PARTS · ONE PARAMETRIC MODEL

Exploded Drawing to Living 3D

A ceiling-hung, frameless-look shower door became an interactive 3D model, a configurator and a step-by-step install guide, so design, procurement and site teams all work from the same model.

~~6.5~~ → **3.2** hours to install one enclosure

3.2 hrs
Install time per enclosure
▼ from about 6.5 hours

−58 %
Snags at handover
▼ per 100 enclosures

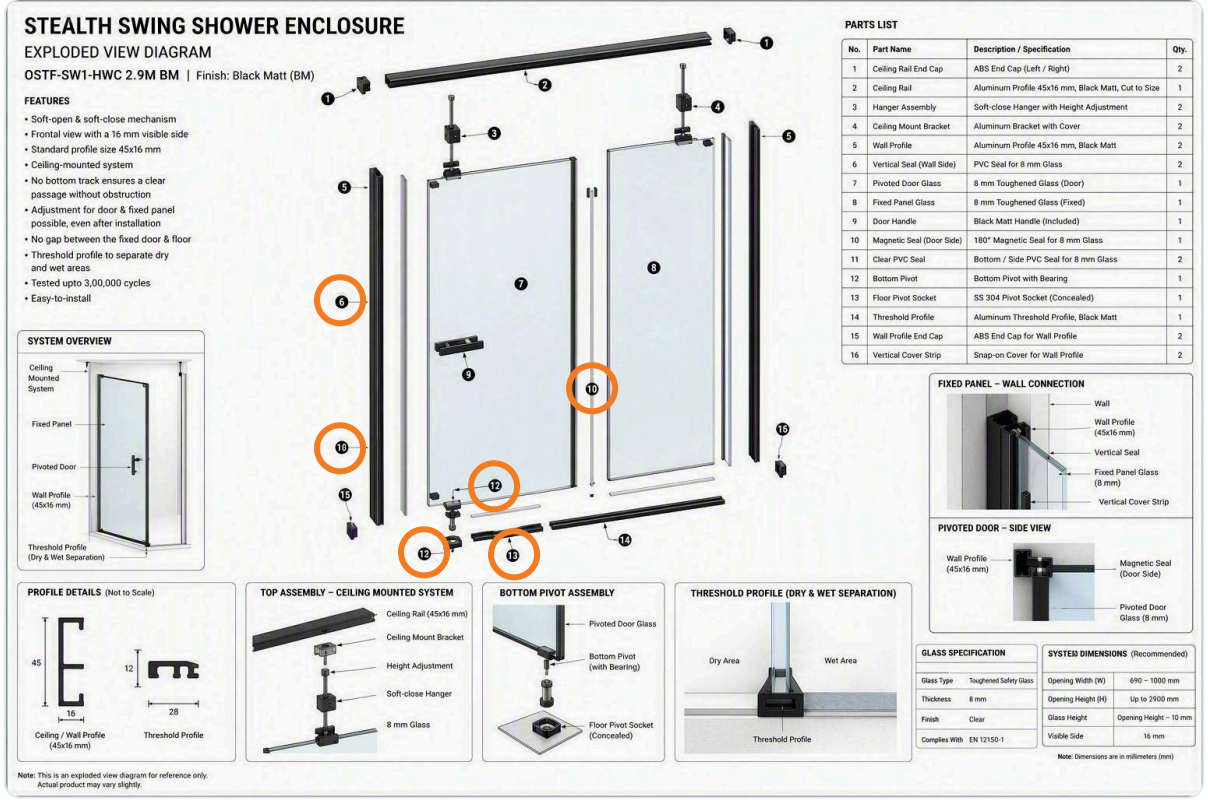
−70 %
Drawing queries from site
▼ installers check the 3D, not email

15 min
Bill of materials per tower
▼ from 2 days of manual take-off

Rings: install hours now (orange) with the earlier 6.5 h inside, on a 9.5 h scale; share of snags and of site queries that remain; one segment per part in the BOM. Figures indicative.

01 Sixteen parts, eight steps, one sequence

The manufacturer's drawing, kept as the reference and linked to the 3D model



No.	Part Name	Description / Specification	Qty.
1	Ceiling Rail End Cap	ABS End Cap (Left / Right)	2
2	Ceiling Rail	Aluminum Profile 45x16 mm, Black Matt, Cut to Size	1
3	Hanger Assembly	Soft-close Hanger with Height Adjustment	2
4	Ceiling Mount Bracket	Aluminum Bracket with Cover	2
5	Wall Profile	Aluminum Profile 45x16 mm, Black Matt	2
6	Vertical Seal (Wall Side)	PVC Seal for 8 mm Glass	2
7	Pivoted Door Glass	8 mm Toughened Glass (Door)	1
8	Fixed Panel Glass	8 mm Toughened Glass (Fixed)	1
9	Door Handle	Black Matt Handle (Included)	1
10	Magnetic Seal (Door Side)	180° Magnetic Seal for 8 mm Glass	1
11	Clear PVC Seal	Bottom / Side PVC Seal for 8 mm Glass	2
12	Bottom Pivot	Bottom Pivot with Bearing	1
13	Floor Pivot Socket	SS 304 Pivot Socket (Concealed)	1
14	Threshold Profile	Aluminum Threshold Profile, Black Matt	1
15	Wall Profile End Cap	ABS End Cap for Wall Profile	2
16	Vertical Cover Strip	Snap-on Cover for Wall Profile	2

GLASS SPECIFICATION	
Glass Type	Toughened Safety Glass
Thickness	8 mm
Finish	Clear
Complies With	EN 12150-1

SYSTEM DIMENSIONS (Recommended)	
Opening Width (W)	690 – 1000 mm
Opening Height (H)	Up to 2900 mm
Glass Height	Opening Height – 10 mm
Visible Side	16 mm

Note: Dimensions are in millimeters (mm)

Pivot and seal parts: where most re-dos came from Every part number became a hotspot to its part in 3D

INSTALL PLAYBACK ON A SITE TABLET

Each step names its parts and its QC check

Part numbers match the drawing · tolerances from the manufacturer's manual (illustrative)

1	**Set out and fix wall profiles** Parts 5, 15	plumb ±1 mm/m
2	**Fix ceiling rail and end caps** Parts 2, 1	level ±1 mm
3	**Set the floor pivot socket** Parts 13	plumb under hanger ±0.5 mm
4	**Mount brackets and fixed panel** Parts 4, 8, 6	plumb ±1 mm/m
5	**Fit hanger and bottom pivot** Parts 3, 12	adjuster at mid-travel
6	**Hang the door, fit the handle** Parts 7, 9	even gap ±1 mm
7	**Fit magnetic and PVC seals** Parts 10, 11	water test: no leak in 5 min
8	**Threshold, cover strips, swing test** Parts 14, 16	10 soft-close cycles

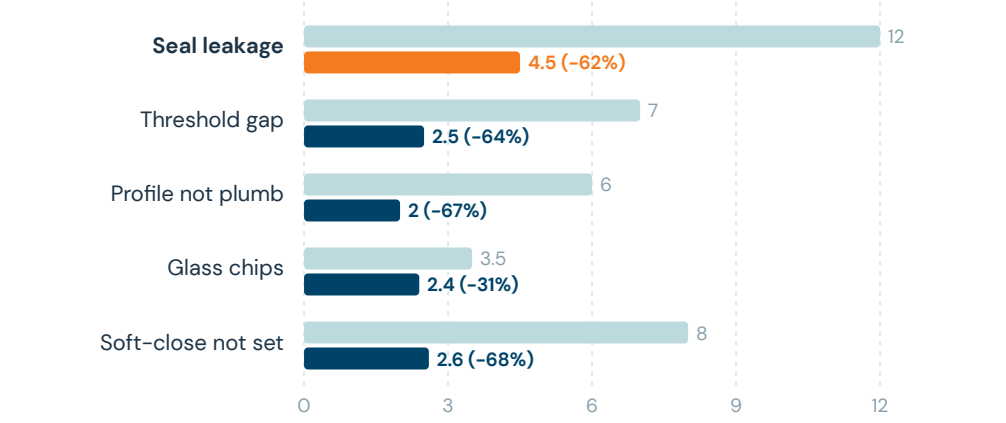
Most snags came from doing steps out of order. Playing the install step by step, with QC signed off at each stage, fixed that.

02 Fewer snags, faster crews

Impact dashboard · figures indicative

Snags per 100 enclosures, by type

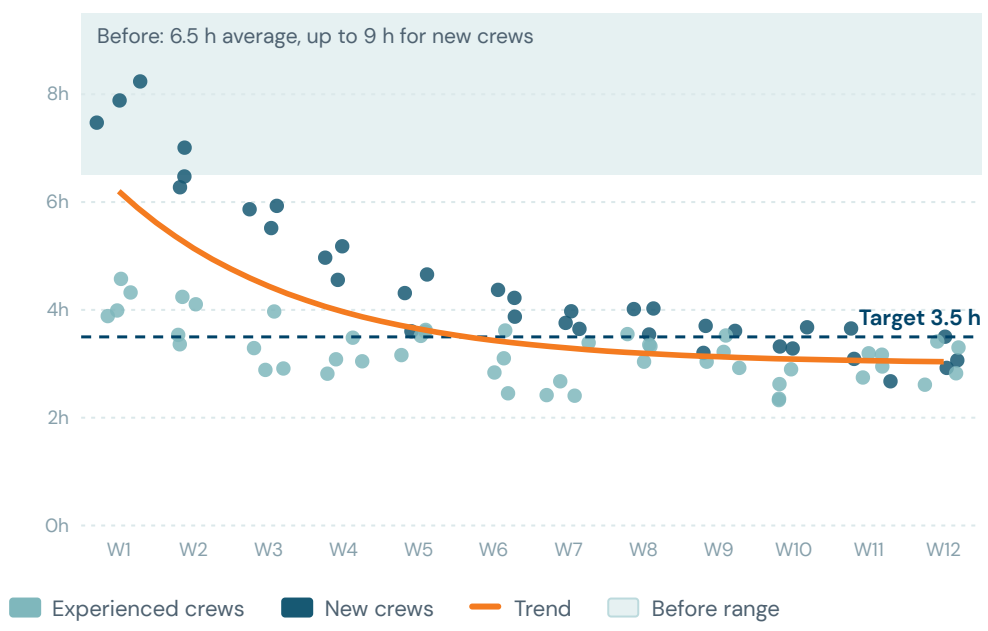
At customer handover · PDF drawing (light) vs 3D studio (navy)



46.0 → **16.8** snags per 100 enclosures, all six types

Crew learning curve

Install time per enclosure by week of rollout · each dot is one install



03 Four pain points, four fixes

A beautiful product, a hard drawing to read on site

BEFORE
Flat drawing, 3D product
Installers had to imagine how hanger, pivot and seals fit from one exploded view printed on A3.

PARAMETRIC 3D MODEL

AFTER
Explode, swing, section
Interactive 3D in any browser, linked part by part to the original drawing.

BEFORE
Slow, uneven installs
About 4 hours for an experienced crew, 8 or more for new crews.

INSTALL PLAYBACK

AFTER
Step by step on a tablet
Crew briefings run from the playback; new crews reached the target time within weeks.

BEFORE
Snags at handover
Door drop, leaking seals and threshold gaps found at walk-throughs, the costliest time to fix.

QC AT EACH STEP

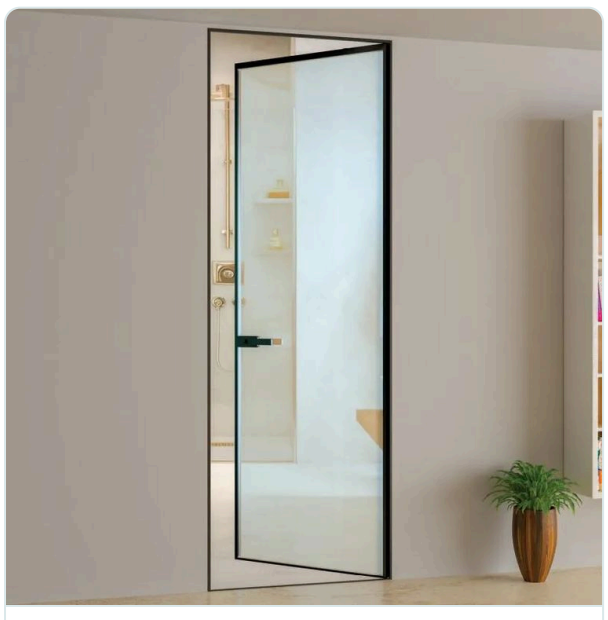
AFTER
Checked as it is built
QC signed off per install step on a tablet, not at handover.

BEFORE
Manual take-offs
Every bathroom size meant a fresh cut list for glass and profiles, by hand, per tower.

CONFIGURATOR + BOM

AFTER
BOM in minutes
Glass height = opening − 10 mm; materials roll up by tower into a CSV for vendor RFQs.

04 The finished product



Black Matt · 8 mm clear glass. A 16 mm visible profile and no bottom track: simple when finished, easy to get wrong on site.

05 Spec to handover

	BEFORE	AFTER
Reference	One static exploded drawing	Interactive 3D, linked to the drawing
Sizing	Cut list worked out per bathroom	Configurator gives glass and profile sizes instantly
Crew briefing	Verbal, from the drawing	Step-by-step install playback on a tablet
Quality	Checked at handover	QC signed off at each install step
Procurement	Manual take-off, about 2 days per tower	BOM roll-up and CSV in minutes
Sales	Photos and finish samples	Finishes shown in 3D at the experience centre

06 How HMC ran it

1 week	**Discover**	Watched three installs; timed each stage and logged every drawing query and snag.
1 week	**Design**	Each of the 16 parts modelled with spec, quantity, install step and QC check.
3 weeks	**Build**	Claude wrote the Three.js model, configurator and BOM logic: one HTML file, offline on a tablet.
1 tower	**Adopt**	Piloted on one tower: briefings from the playback, QC per step.
∞ ongoing	**Scale**	Sliding, walk-in and fixed screens added; BOM roll-up feeds RFQs by tower.

What transfers to any projects team

01
Model the product, not the picture
A parametric model answers questions a drawing cannot: what fits, in what order, at what size.

02
Put the sequence on site
Most snags came from doing steps out of order. Playing the install step by step fixed that.

03
One model, many teams
Design, procurement, site and sales now all use the same model, so they also agree on the details.